

Lab Report -01

Group- 04

Section- 27-(a)

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Lab Experiment -1

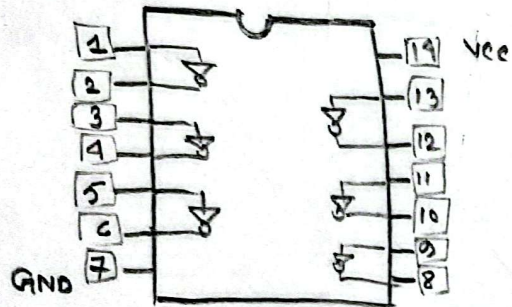
Experiment name: Familiarization of Fundamental
Logic Gates.

Required component's:

1. AND (IC-7408)
2. OR (IC-7432)
3. NOT (IC-7404)
4. NAND (IC-7400)
5. NOR (IC-7402)
6. XOR (IC-7486)
7. XNOR (IC-4077)

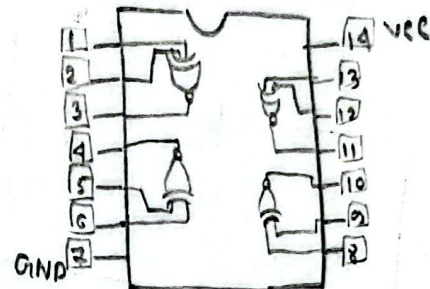
Experimental setup:

NOT



7404

X-NOR



4077

NOT



Input	Output
0	1
1	0

XNOR



Input	Output
0 0	1
1 0	0
0 1	0
1 1	1

Majm
22/02/26

LAB EXPERIMENT-02

Experiment name: Universal Gates and Applications
of Boolean algebra.

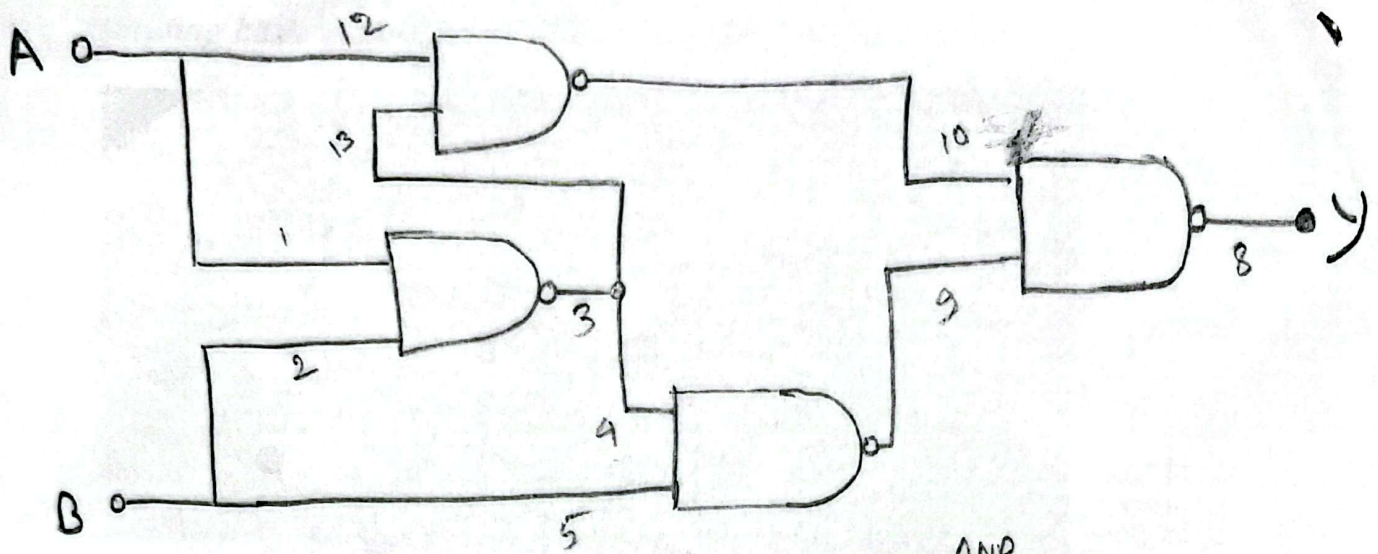
Required component's:

1. IC 7400 $\times$ 1 (NAND gate)

2. IC 7402 $\times$ 2 (NOR gate)

Experimental Status:

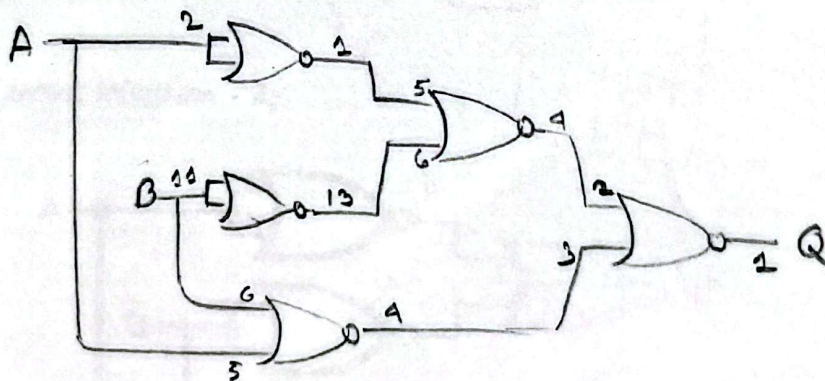
NAND:



ANP

Input	Output
0 0	0
1 0	0
0 1	0
1 1	1

X-OR



XOR

Input	Output
0 0	0
1 0	1
0 1	1
1 1	0

Majid
22/12/20

LAB EXPERIMENT-3

Experiment Name: Parity Generator and checker;

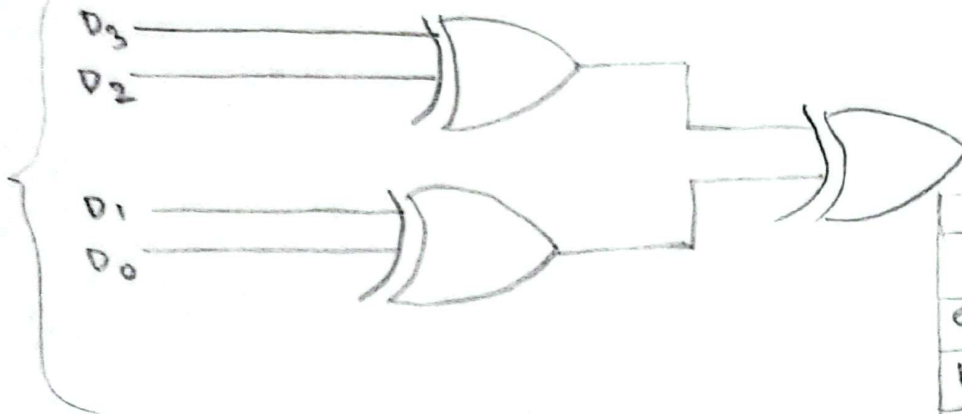
Required Component's :

1. IC 7486 x 1

Experimental setup :

(Group-4)
[Experiment-3]

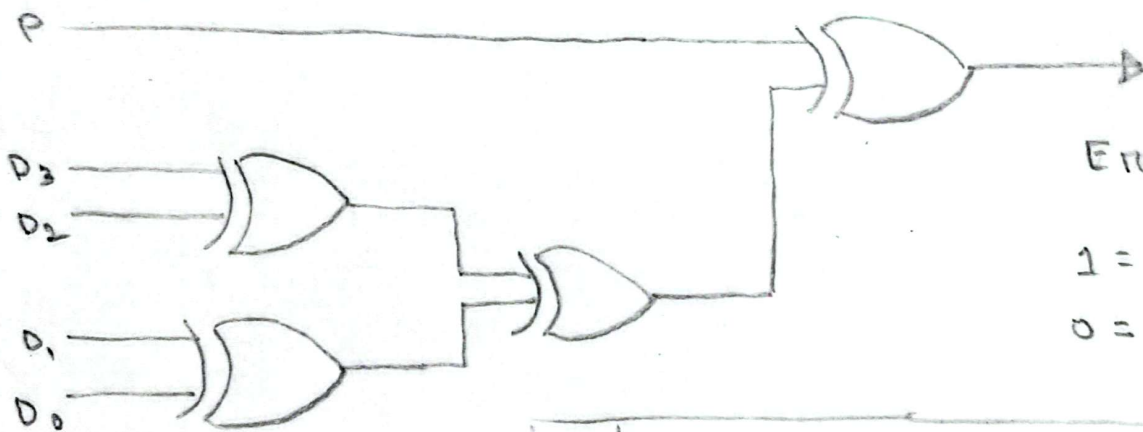
original data



Parity Generator

Parity(P)

	Data				Parity
	D ₃	D ₂	D ₁	D ₀	
a	1	0	0	1	0
b	0	0	0	1	1
c	1	1	1	1	0
d	0	0	0	0	0



Error (E)

1 = Error

0 = No Error

	Data					Error
	Parity	D ₃	D ₂	D ₁	D ₀	
a.	1	1	0	0	1	1
b.	0	0	0	0	1	1
c.	0	1	1	1	1	0
d.	1	0	0	0	0	1

Parity Bit Checker

Result:

Complete both of the following truth tables.

Truth Table for Parity Bit Generator:

	Data				Parity
	D ₃	D ₂	D ₁	D ₀	
a.	1	0	0	1	0
b.	0	0	0	1	1
c.	1	1	1	1	0
d.	0	0	0	0	0

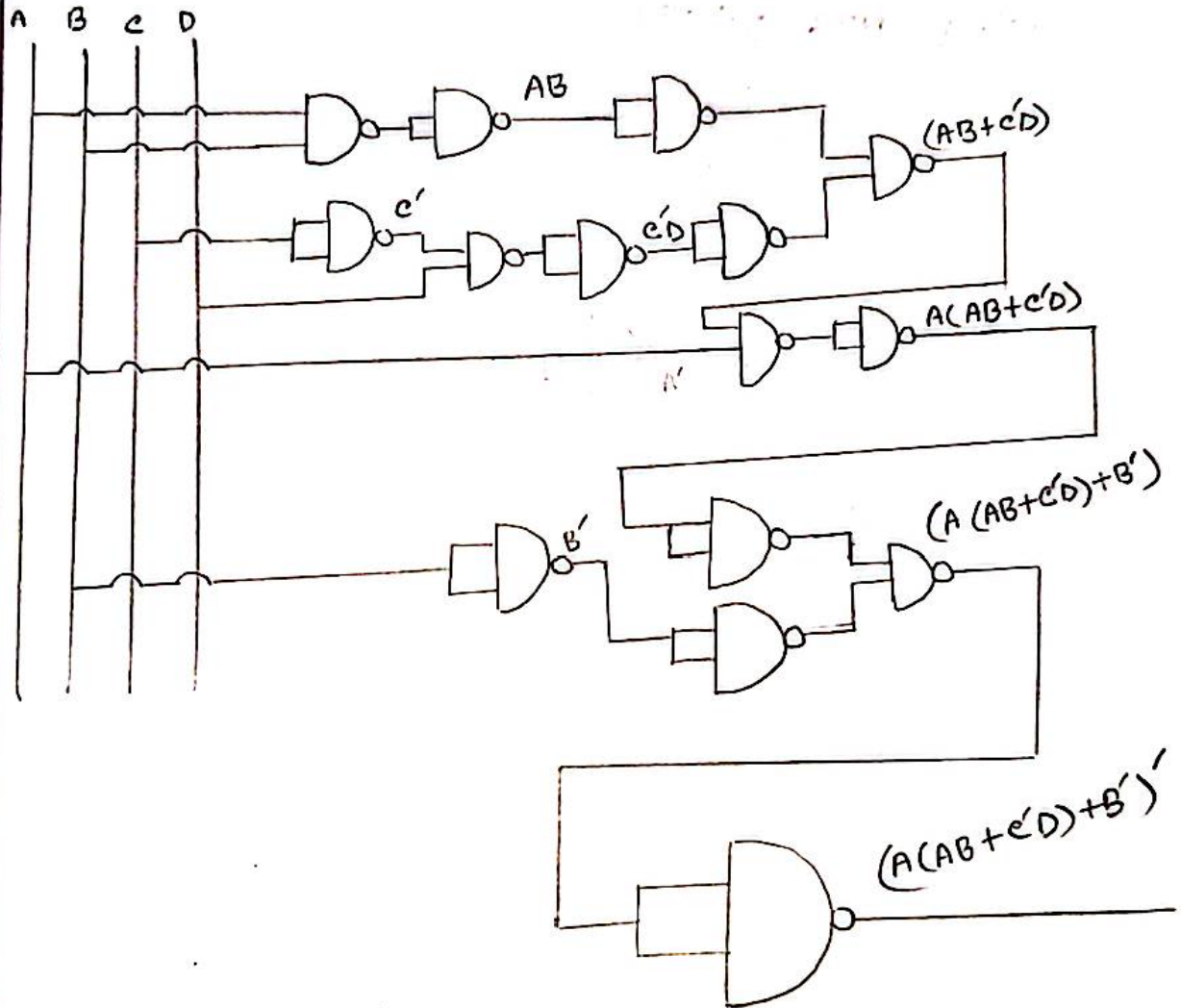
Truth Table for Parity Bit Checker:

	Data					Error
	Parity	D ₃	D ₂	D ₁	D ₀	
a.	1	1	0	0	1	1
b.	0	0	0	0	1	1
c.	0	1	1	1	1	0
d.	1	0	0	0	0	1

[Discussion]

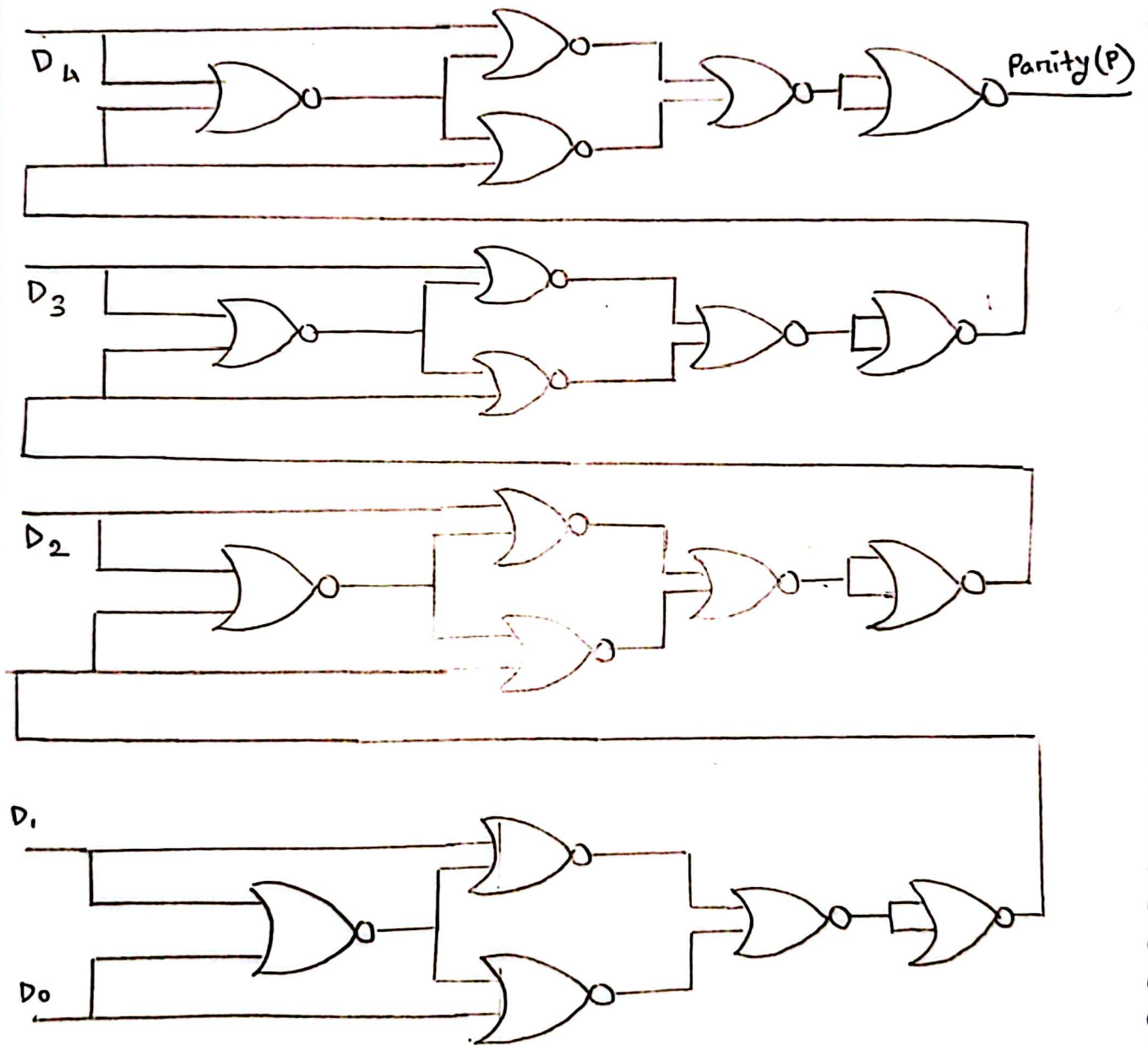
[Ans to the question no 1]

$$F = (A(AB + C'D) + B')'$$



[Ans to the question no 2]

5-bit odd parity checker generator :
(using NOR gates)



5-bit odd parity checker:
(using NOR gates)

